A Geometric Identity for the Fine-Structure Constant α: From the 4D Unit Ball Volume and its Cube-Packing Deficit

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Main Result: a self-consistent algebraic identity predicts α to 8.7 ppb Two elementary 4D geometric quantities — $N(1) = 137$ (the count of integer-centred unit cubes fully contained in a 4D ball of radius 3, by direct enumeration) and $V_4(1) = \pi^2/2$ (the 4D unit ball volume by standard formula) — satisfy the self-consistent equation

$$\alpha^{-1} = N(1) + V_4(1)\,\alpha = 137 + \tfrac{\pi^2}{2}\alpha \quad \Longleftrightarrow \quad \tfrac{\pi^2}{2}\alpha^2 + 137\alpha - 1 = 0$$

whose positive root $\alpha = 7.29735194 \times 10^{-3}$ matches the CODATA 2018 observed value $7.29735257 \times 10^{-3}$ with relative error $\mathbf{8.7 \times 10^{-8}}$ (**8.7 ppb**, about 1/3000 the deviation of Eddington-style integer-fitting at 2.6×10^{-4}).

Decisive differences from the Eddington trap (1) Not pure integer-fitting; a **self-consistent equation containing α itself**. (2) Both the integer 137 and the correction coefficient $\pi^2/2$ are **derived directly and independently from 4D geometry** (not borrowed). (3) The perturbation expansion $1 = 137\alpha + V_4(1)\alpha^2$ is structurally analogous to QFT renormalisation. (4) Candidate physical mechanisms are proposed from independent thought-experiment work.

Caveat This paper does not claim a first-principles derivation of α. What is presented is an observed numerical structure. Whether that reflects geometric necessity or numerical coincidence is left as an open problem.

$N(1) = 137$: direct enumeration The condition $\sum_i (|c_i| + 1/2)^2 \le 9$ for an integer-centred unit cube to be fully contained in the 4D ball of radius $R = 3$. Enumerating by absolute-value patterns $(s_1, \ldots, s_4)$:

Pattern	Cube count
$(0,0,0,0)$	1
$(1,0,0,0)$	8
$(1,1,0,0)$	24
$(1,1,1,0)$	32
$(1,1,1,1)$	16
$(2,0,0,0)$	8
$(2,1,0,0)$	48
Total	**137**

$(2,1,1,0)$ and beyond fail $\sum s_i^2 + \sum s_i \le 8$. Fully verifiable by hand.

$V_4(1) = \pi^2/2$: standard formula $V_n(R) = \pi^{n/2}/\Gamma(n/2+1) \cdot R^n$ at $n = 4, R = 1$: $V_4(1) = \pi^2/\Gamma(3) = \pi^2/2 \approx 4.9348$. Coincides with the Planck-unit 4+1D BH entropy $S_{BH}(1) = \pi^2/2$ (cross-reference).

Perturbative structure The equivalent form $1 = 137\alpha + (\pi^2/2)\alpha^2$ is a power expansion in α: tree-level 137α + first-order correction $V_4(1)\alpha^2$ summing to unity. Analogous to QFT renormalisation.

Inside / outside decomposition The R=3 ball has volume $V_4(3) = 81\pi^2/2 \approx 399.72$. Of this, 137 cubes occupy the interior (34%); the boundary gap is **66%**. Boundary-dominated.

- 137α: inside, 137 geometric contact points
- $V_4(1)\alpha^2$: outside, boundary self-energy correction

Numerical reproduction (calculator-level)

$V_4(1) = \pi^2/2$	≈ 4.9348022
$N(1)$	$= 137$
$D = 137^2 + 2\pi^2$	≈ 18788.7392
$\sqrt{D}$ (high precision)	≈ 137.0720220
α (high precision)	$\approx 7.29735194 \times 10^{-3}$
α^{-1} predicted	≈ 137.0360110
α^{-1} (CODATA 2018)	$= 137.0359991\ldots$
relative error	$\mathbf{8.7 \times 10^{-8}}$ (8.7 ppb)

Candidate mechanisms

- A: vacuum polarisation (QED virtual pairs)
- B: boundary-mode self-correlation
- C: beat coupling with longitudinal Higgs

Not exclusive; possibly facets of the same phenomenon.

Open problems

- Geometric identification of the 0.03% residual
- Why $N(1)$ corresponds to the electron (R=3 privilege)
- First-principles derivation of mechanism
- Generation mass hierarchy is out of scope

What this paper does *not* claim

- First-principles derivation of α
- Physical spacetime is 4+1-dimensional
- Replacement for Eddington-style predictions
- New observable signatures

Paper DOIs Concept: 10.5281/zenodo.19869266
v1: 10.5281/zenodo.19869267

Note article (English) https://note.com/kiharanoriaki/n/n01d98237ddae

Note article (Japanese) https://note.com/kiharanoriaki/n/n19cc13927c51

Zenn article (Japanese, technical exposition) https://zenn.dev/noriaki_kihara/articles/alpha-identity-4d-geometry

Zenodo (Paper PDF, LaTeX, MD, JA/EN bilingual) https://zenodo.org/records/19869267

Source code (GitHub): https://github.com/WurabeSeiji/ai-chat-logs-open/tree/main/BH/papers/paper7_alpha Related: BH Thermodynamics Programme Zenn / note EN